

# SCTS - MACHINE LEARNING SYLLABUS

Course Code: CS-401 | Core Module | Credits: 4

Course Overview: Machine Learning algorithms, statistical modeling, neural networks.

Unit 1: Linear Regression, Logistic Regression, Cost Functions, and Regularization.

Unit 2: Decision Trees, Random Forests, Support Vector Machines.

Unit 3: Deep Neural Networks, Backpropagation, Convolutional Networks.

Unit 4: Unsupervised Learning, K-Means Clustering, PCA Dimensionality Reduction.

Reference Books:

1. Pattern Recognition and Machine Learning - Christopher Bishop
2. Hands-On Machine Learning - Aurelien Geron